

# Tung Vo

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## EDUCATION

**University of St. Thomas** - Houston, Texas

*Expected May 2028*

Bachelor of Science in Applied Mathematics; Bachelor of Arts in Mathematics - GPA: 3.91 / 4.00

**Lone Star College** - Houston, Texas

*May 2025*

Honors Associate of Science, Summa Cum Laude - Distinguished Global Scholar (International Study)

## RESEARCH INTERESTS

Enumerative combinatorics and parking functions · Combinatorial optimization and integer programming · Graph theory and zero forcing · Quantum circuit optimization

## RESEARCH EXPERIENCE

**Undergraduate Researcher**, Rice University (CMOR Department) - Houston, TX

*Spring 2026 - Present*

Research Advisor: Dr. Illya Hicks · Project Lead: Saber Dinpazhouh

- Designing an integer-programming heuristic for the qubit routing problem, mapping logical qubits onto hardware-constrained physical qubits while minimizing the SWAP gates inserted during compilation.
- Benchmarking the heuristic against established routing methods (SABRE, TKET) on gate count and circuit depth to evaluate compilation efficiency.
- Scaling the heuristic to circuits of 14+ qubits, where exact integer-programming solvers become intractable and fail to return solutions within practical time limits.

**Undergraduate Researcher**, University of St. Thomas - Houston, TX

*Fall 2025 - Present*

Research Advisor: Dr. Mary Flagg

- Defining and analyzing competitive zero forcing, a two-player graph coloring game that extends the classical zero-forcing color-change rule to an adversarial setting.
- Conducting theoretical analysis and computational modeling of vertex connectivity and forcing dynamics across multiple graph families.
- Presenting joint findings at the University of St. Thomas Annual Research Symposium (2026).

**Undergraduate Researcher**, University of Puerto Rico - Ponce - Ponce, PR

*Summer 2026*

Research Advisor: Dr. Pamela Harris and Dr. Fernando Piñero

- Proving enumeration and structural results on weakly increasing and decreasing Fubini rankings, unit interval parking functions, and Cayley permutations under Dr. Pamela Harris.
- Characterizing the intersections of these three combinatorial families, including an if-and-only-if structural classification of  $C_n \cap UPF_n$ .
- Contributed to the mathematics and content of the research manuscript.

**Undergraduate Researcher**, Lone Star College Honors College - Houston, TX

*Fall 2024*

Research Advisor: Dr. Kelly Phillips

- Examined how U.S. lottery-draft and Viet Cong propaganda recruitment tactics shaped soldier desertion rates (1950-1975).
- Analyzed primary-source propaganda posters and draft records, finding race and education level significantly affected draft likelihood.

**Undergraduate Researcher**, Lone Star College Honors College - Houston, TX

*Spring 2024*

Research Advisors: Dr. Sean Tiffée, Dr. Rebecca Howard, Dr. Peggy Lambert, Prof. Casey Garcia

- Authored an autoethnography of competing at National Model UN-New York, applying triangulation and content analysis to examine how intercultural communication and public speaking drive leadership in large delegate settings.

**Undergraduate Researcher**, Lone Star College Honors College - Houston, TX

Spring 2024

Research Advisor: Dr. Dana Van De Walker

- Applied a Most Similar System Design (MSSD) case-study framework to compare air pollution and life expectancy in the U.S. and Chad.
- Identified socioeconomic status, infant mortality, healthcare access, and reforestation investment as confounding drivers beyond pollution alone.

## PUBLICATIONS

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### Research Articles

(1) *Cayley Permutations, Fubini Rankings, and Unit Interval Parking Functions*

Eban Barret, Melissa Beerbower, Jennifer Elder, Pamela E. Harris, Maritza Sanchez, Tung Vo, and Cyrus Young.  
In preparation.

## PRESENTATIONS

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Vo, T., Barrett, E., & Sanchez, M. "*Enumerating Weak Order Fubini Rankings and Unit Interval Parking Functions.*"  
Young Mathematician Conference, Ohio State University, Columbus, OH, 2026.

Vo, T., Cotsoradis, S., & Rutherford, C. "*Competitive Zero Forcing: A Novel Two-Player Graph Coloring Game.*"  
Annual Research Symposium, University of St. Thomas, Houston, TX, 2026.

Vo, T. "*A Systemic Approach to Understanding the Natural World.*" Rice University Environmental Data Academy  
Poster Session, Houston, TX, 2025.

Vo, T. "*A Systemic Approach to Understanding the Natural World.*" Spring Honors & International Education  
Conference, Houston, TX, 2025.

Vo, T. "*The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War.*"  
World History Association of Texas Annual Conference, Commerce, TX, 2025.

Vo, T. "*The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War.*" Fall  
Honors & International Education Conference, Houston, TX, 2024.

Vo, T. "*The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War.*"  
International Education Biennial Conference, Houston, TX, 2024.

Vo, T. "*Life Expectancy and Air Pollution: A Comparative Analysis of the United States and Chad.*" Spring Honors &  
International Education Conference, Houston, TX, 2024.

## OUTREACH

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**Finalist**, CASE Program - Amideast Education Abroad Connect - Tunis, Tunisia

Summer 2025

- Selected for an 8-day civil-society program engaging regional leaders on civic engagement, sustainability, and global justice.

**Campus Representative**, LSC Honors College Student Advisory Board

2024 - 2025

- Advised the Associate Vice Chancellor on Honors College programming and system-wide student activities.

**System Liaison Emissary**, LSC Honors & International Education

2024 - 2025

- Promoted Honors College, Rice Take Flight, Model UN, and scholarship programs; mentored the incoming liaison as Emeritus.

**Press Team Member & Statesman**, Texas Boys State (American Legion)

Summer 2022

- Selected as 1 of 2 students from a class of 800; covered mock House proceedings and ran social-media campaigns.

**AWARDS & HONORS**

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|----------------------------------------------------------------------------------------|-------------|
| Dean’s List, University of St. Thomas (Fall 2025, Spring 2026)                         | 2025 - 2026 |
| Hispanic-Serving Institution STEM (Dunn) Endowed Scholarship, University of St. Thomas | 2026        |
| Monaghan Excellence Scholarship, University of St. Thomas                              | 2025 - 2027 |
| Distinguished Global Scholar Study Abroad Scholarship, Lone Star College               | 2025        |
| President’s List, Lone Star College (All semesters)                                    | 2023 - 2025 |
| Global Scholar Award & Language Scholarship, Lone Star College                         | 2023 - 2025 |
| Best in Committee & Outstanding Delegation, National Model UN, New York                | 2024        |
| National Speech & Debate Association Academic All-American                             | 2022        |

**SKILLS**

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Environmental Programming & Tools: Python, R, spreadsheets, integer programming, LaTeX.  
Languages: Vietnamese (native speaker), English (fluent)